

Ratio, Proportion and Rates of Change

GCSE Maths · Worked Solutions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

Write the ratio 12 : 18 in its simplest form.

- 1 Find the highest common factor of 12 and 18, which is 6.
- 2 Divide both sides by 6: $12 \div 6 = 2$ and $18 \div 6 = 3$

Answer: 2 : 3

2 [2 marks]

Share £60 in the ratio 2 : 3

- 1 Add the parts of the ratio: $2 + 3 = 5$ parts altogether.
- 2 One part is worth $£60 \div 5 = £12$
- 3 2 parts = $2 \times 12 = £24$ and 3 parts = $3 \times 12 = £36$
- 4 Check: $24 + 36 = 60$ ✓

Answer: £24 and £36

3 [2 marks]

A recipe for 4 people uses 200 g of rice.

How much rice is needed for 10 people?

- 1 Find the amount for one person first: $200 \div 4 = 50$ g per person.
- 2 Then multiply by 10: $50 \times 10 = 500$ g

Answer: 500 g

4 [2 marks]

Work out 15% of 240

- 1 $10\% \text{ of } 240 = 240 \div 10 = 24$
- 2 5% is half of 10%, so $5\% \text{ of } 240 = 12$
- 3 $15\% = 10\% + 5\% = 24 + 12 = 36$

Answer: 36

5

[3 marks]

A map is drawn to a scale of 1 : 25 000

Two villages are 6 cm apart on the map.

Work out the real distance between the villages, in kilometres.

1 The scale means 1 cm on the map represents 25 000 cm in real life.

2 6 cm represents $6 \times 25\,000 = 150\,000$ cm

3 Convert to metres: $150\,000 \div 100 = 1500$ m

4 Convert to kilometres: $1500 \div 1000 = 1.5$ km

Answer: 1.5 km

6

[3 marks]

(a) Increase £48 by 25%

(b) Decrease 90 kg by 30%

1 (a) 25% of 48 = $48 \div 4 = 12$

2 Increase means add on: $48 + 12 = £60$

3 (b) 10% of 90 = 9, so 30% of 90 = 27

4 Decrease means take away: $90 - 27 = 63$ kg

5 (A quicker method is to multiply by 1.25 and by 0.7.)

Answer: (a) £60 (b) 63 kg

7

[3 marks]

A 500 g box of cereal costs £1.80

An 800 g box of the same cereal costs £2.72

Which box is better value for money? You must show your working.

1 Compare the two boxes using the same measure — the cost of 1 gram.

2 Small box: $180\text{p} \div 500 = 0.36\text{p}$ per gram

3 Large box: $272\text{p} \div 800 = 0.34\text{p}$ per gram

4 The large box costs less per gram, so it is better value.

Answer: The 800 g box (0.34p per gram against 0.36p per gram)

8

[3 marks]

A car travels 150 km in 2 hours 30 minutes.

Work out the average speed of the car in km/h.

1 The time must be in hours to get an answer in km/h.

2 30 minutes = 0.5 hours, so the time is 2.5 hours.

3
$$\text{Speed} = \frac{\text{distance}}{\text{time}} = \frac{150}{2.5}$$

4 $150 \div 2.5 = 60$

Answer: 60 km/h

9

[4 marks]

In a class the ratio of boys to girls is 3 : 5

There are 12 more girls than boys.

Work out the total number of students in the class.

- 1 The difference between the parts of the ratio is $5 - 3 = 2$ parts.
- 2 Those 2 parts are worth 12 students.
- 3 So 1 part = $12 \div 2 = 6$ students.
- 4 The total is $3 + 5 = 8$ parts, so the class has $8 \times 6 = 48$ students.
- 5 Check: boys = $3 \times 6 = 18$, girls = $5 \times 6 = 30$, and $30 - 18 = 12$ ✓

Answer: 48 students

10

[4 marks]

Here is a recipe for 12 biscuits.

Flour	Butter	Sugar
150 g	80 g	60 g

Kim has 400 g of flour, 200 g of butter and 200 g of sugar.

What is the greatest number of biscuits Kim can make?

- 1 Work out how many batches of 12 each ingredient would allow.
- 2 Flour: $400 \div 150 = 2.66\dots$ batches
- 3 Butter: $200 \div 80 = 2.5$ batches
- 4 Sugar: $200 \div 60 = 3.33\dots$ batches
- 5 The butter runs out first, so it limits her to 2.5 batches.
- 6 $2.5 \times 12 = 30$ biscuits.

Answer: 30 biscuits (butter is the limiting ingredient)

Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

Write the ratio 4 : 10 in the form 1 : n

- 1 To make the left-hand side 1, divide both sides by 4.
- 2 $4 \div 4 = 1$ and $10 \div 4 = 2.5$

Answer: 1 : 2.5

2

[2 marks]

Express 45 minutes as a percentage of 3 hours.

- 1 Put both times into the same units: 3 hours = 180 minutes.
- 2 Write the fraction: $\frac{45}{180} = \frac{1}{4}$
- 3 Convert to a percentage: $\frac{1}{4} \times 100 = 25\%$

Answer: 25%

3

[3 marks]

In a sale, the price of a jacket is increased by 20%.

The new price is £54

Work out the original price of the jacket.

- 1 This is a reverse percentage — do not simply take 20% off £54.
- 2 The original price is 100%; after a 20% increase the new price is 120% of it.
- 3 So $120\% = £54$, which means the multiplier used was 1.2
- 4 Original price = $54 \div 1.2 = £45$
- 5 Check: $45 \times 1.2 = 54 \checkmark$

Answer: £45

4

[3 marks]

Ravi invests £2000 in an account paying 3% compound interest per year.

Work out the value of the investment after 4 years.

Give your answer to the nearest penny.

- 1 Compound interest multiplies by the same factor every year.
- 2 The multiplier for a 3% increase is 1.03
- 3 After 4 years the value is 2000×1.03^4
- 4 $1.03^4 = 1.12550881$
- 5 $2000 \times 1.12550881 = 2251.01762$, which is £2251.02 to the nearest penny.

Answer: £2251.02

5

[4 marks]

y is inversely proportional to x^2 .

When $x = 2$, $y = 9$

(a) Find a formula for y in terms of x .

(b) Work out y when $x = 6$

(c) Work out the positive value of x when $y = 4$

- 1 (a) Inversely proportional to x^2 means $y = \frac{k}{x^2}$

- 2 Substitute $x = 2$ and $y = 9$: $9 = \frac{k}{4}$, so $k = 36$

- 3 The formula is $y = \frac{36}{x^2}$

- 4 (b) $y = \frac{36}{6^2} = \frac{36}{36} = 1$

- 5 (c) $4 = \frac{36}{x^2}$, so $4x^2 = 36$ and $x^2 = 9$

- 6 The positive root is $x = 3$

Answer: (a) $y = \frac{36}{x^2}$ (b) 1 (c) 3

6

[4 marks]

A bag contains red counters and blue counters in the ratio 5 : 3

10 more blue counters are put into the bag.

The ratio of red to blue counters is now 5 : 4

Work out the total number of counters now in the bag.

- 1 Let the original numbers be $5x$ red and $3x$ blue.
- 2 After adding 10 blue counters the ratio is $5x : (3x + 10) = 5 : 4$
- 3 Cross-multiply: $4 \times 5x = 5 \times (3x + 10)$, so $20x = 15x + 50$
- 4 $5x = 50$, so $x = 10$
- 5 Red = $5 \times 10 = 50$ and blue = $3 \times 10 + 10 = 40$
- 6 Total = $50 + 40 = 90$. Check: $50 : 40$ simplifies to $5 : 4$ ✓

Answer: 90 counters

7

[4 marks]

Two solids are mathematically similar.

The smaller solid has surface area 50 cm^2 and volume 40 cm^3 .

The larger solid has surface area 200 cm^2 .

Work out the volume of the larger solid.

- 1 For similar solids the area scale factor is the square of the length scale factor.
- 2 Area scale factor = $\frac{200}{50} = 4$
- 3 So the length scale factor is $\sqrt{4} = 2$
- 4 The volume scale factor is the cube of the length scale factor: $2^3 = 8$
- 5 Volume of larger solid = $40 \times 8 = 320 \text{ cm}^3$

Answer: 320 cm^3

8

[4 marks]

A block of metal has mass 2.4 kg and volume 300 cm^3 .

(a) Work out the density of the metal in g/cm^3 .

(b) A second block of the same metal has volume 750 cm^3 . Work out its mass in kilograms.

- 1 **(a)** The units g/cm^3 tell you the mass must be in grams: $2.4 \text{ kg} = 2400 \text{ g}$
- 2 Density = $\frac{\text{mass}}{\text{volume}} = \frac{2400}{300} = 8 \text{ g/cm}^3$
- 3 **(b)** Same metal means the same density, so rearrange to mass = density $\times$ volume.
- 4 Mass = $8 \times 750 = 6000 \text{ g}$
- 5 Convert back: $6000 \div 1000 = 6 \text{ kg}$

Answer: (a) 8 g/cm^3 (b) 6 kg

9

[4 marks]

A lorry drives 120 km at an average speed of 80 km/h.

It then drives a further 60 km at an average speed of 40 km/h.

Work out the average speed for the whole journey.

1 You cannot just average 80 and 40 — the two stages take different amounts of time.

2 Find each time using $\text{time} = \frac{\text{distance}}{\text{speed}}$

3 First stage: $120 \div 80 = 1.5$ hours

4 Second stage: $60 \div 40 = 1.5$ hours, so the total time is 3 hours.

5 Total distance = $120 + 60 = 180$ km

6 Average speed = $\frac{180}{3} = 60$ km/h

Answer: 60 km/h

10

[5 marks]

The population of a town is 40 000.

The population is falling by 4% each year.

(a) Work out the population after 5 years, to the nearest whole number.

(b) Find the least number of complete years it takes for the population to fall below 30 000.

1 (a) A 4% fall means multiplying by $1 - 0.04 = 0.96$ each year.

2 After 5 years: $40\,000 \times 0.96^5$

3 $0.96^5 = 0.8153726976$, so the population is $40\,000 \times 0.8153726976 = 32\,614.9\dots$

4 To the nearest whole number that is 32 615 people.

5 (b) Try successive years until the population drops below 30 000.

6 After 7 years: $40\,000 \times 0.96^7 = 40\,000 \times 0.751447\dots = 30\,057.9$ — still above 30 000.

7 After 8 years: $40\,000 \times 0.96^8 = 40\,000 \times 0.721389\dots = 28\,855.6$ — below 30 000.

8 So it takes 8 complete years.

Answer: (a) 32 615 (b) 8 years